Computational Neuroscience — Module 07: Communication — Study Questions

1. Name five key brain regions in the social brain network and describe each one's function.
2. What is the difference between the mirror neuron system and the mentalizing network?
3. How does the mirror neuron system support understanding "what" someone is doing?
4. How does the mentalizing network support understanding "why" someone is doing something?
5. Where is the temporoparietal junction (TPJ) and what is its role in Theory of Mind?
6. What is the fusiform face area? Why is face recognition so important for social cognition?
7. Describe the language processing hierarchy: auditory cortex → Wernicke's → Broca's → frontal.
8. What is the N400 ERP component? What triggers it?
9. What is the P600 ERP component? How does it differ from the N400?
10. How are N400 and P600 interpreted as prediction errors at different levels of the language hierarchy?
11. What role does oxytocin play in social cognition?
12. How does oxytocin relate to precision weighting in Active Inference?
13. How might autism be understood as a difference in social precision weighting?
14. What does it mean to "over-weight sensory precision and under-weight social priors"?
15. How does the amygdala contribute to social cognition?
16. What brain regions are active when you infer someone's emotions from their facial expression?
17. How does the brain process sarcasm? Which regions must collaborate?
18. What is prosody and how does the superior temporal sulcus process it?
19. How does social isolation affect the social brain network?
20. Compare the computational neuroscience view of communication with the cognitive science view (Unit 1, Module 07).